TEAM DENOVO · IBAB

BIET

Budget Impact Estimation Tool

Five-year payer budget forecasts for early-stage drug programmes.

THE DEMONSTRATION MODEL, END TO END

8,000,000
covered lives, Germany

↓

181,764
eligible patients, year 1

↓

€296.6M
net budget impact, 5 years

early-stage-bia.onrender.com

10-minute presentation · Round 2

“What will this do to my budget?”

It is the first question a payer asks, and the one an early-stage team is least equipped to answer. Answering it once takes weeks of bespoke spreadsheet work. Answering it again takes another few weeks.

How it is answered today

- Built from scratch in Excel, for every product and every market.
- Formulas sit in cells no reviewer can trace, so the answer is accepted or rejected whole.
- Changing one assumption means rebuilding, so the alternatives never get tested.

Who carries the cost of that

- **HEOR & market access**
  Defend every assumption to a payer with no audit trail to point at.
- **Payers & insurers**
  Receive a single number they cannot interrogate before committing budget.
- **Patients**
  Wait while the same analysis is rebuilt market by market.

Budget impact answers affordability: what it costs the budget. Cost-effectiveness asks a different question, whether the drug is good value. BIET answers the first, and says so.

Six inputs in. A five-year payer forecast out.

One guided flow holds the whole model. Change any input and every figure moves with it, so the analysis is something you interrogate during the meeting rather than after it.

AFFORDABILITY

€0.13

per covered member per month. The same fact as €296.6M over five years, in the unit a payer negotiates on.

WHAT DECIDES IT

New drug annual price

Ranked first of ten assumptions, each moved ±20%. Evidence spend goes where it changes the answer.

1. **The engine runs in the browser**
   No network call sits between an input and its result. A five-year model takes minutes to build and seconds to re-run, against weeks of spreadsheet work.

2. **Evidence sits beside the model**
   PubMed, ClinicalTrials.gov, openFDA, WHO GHO and the World Bank are searched inside the same tool, so an assumption and its source stay together.

3. **Every figure traces back**
   A methodology view follows the calculation from covered population to headline without anyone opening the code.

Drug X, an early-stage GLP-1, in Germany

Type 2 diabetes. A statutory health insurer's view, over five years, 2026 to 2030.

1 **Therapy area**

2 **Population**

3 **Current care**

4 **Intervention**

5 **Uptake**

6 **Outcomes**

Results
7 views over one calculation

THREE THINGS TO WATCH

The funnel

8,000,000 covered lives narrow to 181,764 eligible patients. Every currency figure in the tool hangs off that number.

One answer, two units

€296.6M over five years is €0.13 per member per month. The second is the one a payer negotiates on.

What would change it

Ten assumptions moved ±20% and ranked by how far the answer swings. It says where to spend the evidence budget.

All figures in the demonstration scenario are illustrative. They are plausible for a walkthrough, not drawn from a validated published source.

One answer, in the unit that decides it

€296.6M

Net budget impact over five years, against current care

€0.13

Per covered member per month in year 1, the negotiating unit

181,764

eligible patients, year 1

67,083

patients treated at peak, year 5

€2,527

cost per treated patient-year, on a €3,200 list price

€613

annual price at which net impact reaches zero

2,480

clinical events avoided, worth €17.96M

€148.3M – €444.9M

five-year impact at half and 1.5× uptake

Demonstration values. Plausible for a walkthrough but not drawn from a validated published source, and every PDF the tool exports says so.

The calculation runs in the browser

One React app holds the entire engine. A thin Express service does only what a browser cannot: reach evidence APIs that block cross-origin calls, hold an LLM key it must not leak, and persist a run.

BROWSER · REACT 18 + VITE · NO NETWORK CALL IN THE CALCULATION PATH

6 input tabs
one model object in React state

biaEngine.js
11 functions, all the arithmetic

7 result views
rendered, never recalculated per tab

PDF + Excel
generated client-side

↓ HTTPS, same origin

RENDER WEB SERVICE · EXPRESS 4 · ONE ORIGIN, NO CALCULATION

Static host
serves the built React bundle

Evidence API
/api/v1/evitrack

LLM proxy
/api/chat, key stays server-side

Run history
/api/model/runs

Optional and external: PostgreSQL 16, 14 tables · PubMed, ClinicalTrials.gov, openFDA, WHO GHO, World Bank · Groq, Gemini

38 unit tests passing · 62 source files · one engine. A second copy on the server disagreed by 16% on identical inputs, so we deleted it.

What we would build next, and why

Framed by what the tool does not do today. These are the four limitations we would remove first, in that order.

1 Probabilistic sensitivity

Today: a one-way tornado, ten parameters moved ±20% one at a time.

Next: Monte Carlo over correlated parameters, so the headline carries a confidence interval instead of a point.

2 Multi-country roll-up

Today: one country per run, one currency, one set of prices.

Next: several markets aggregated into a regional total, which is how global teams actually plan.

3 Subgroups with their own parameters

Today: a subgroup narrows the population but inherits the parent's costs and outcomes.

Next: independent cost and outcome parameters, so subgroup economics stop being an approximation.

4 Learned uptake curves

Today: the uptake curve is typed in year by year.

Next: curves fitted from comparable launches, because uptake moves the answer more than price does.

Team Denovo

Institute of Bioinformatics and Applied Biotechnology.

SR

Sumithra Raju

Model & calculation engine

The funnel, cost build-up, projection and break-even search.

NA

Nakulan Anandhan

Interface & experience

The six guided input tabs and the seven result views over them.

VD

Vineeth Devadiga

Evidence module

EviTrack: five external sources behind one search and one workspace.

AK

Atharva Kadam

Backend & data

Express API, the 14-table PostgreSQL schema and Excel import.

NK

Nilesh Kamble

Validation & deployment

The test suite, the Render deployment and the PDF and Excel exports.

Questions

Thank you.

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Live

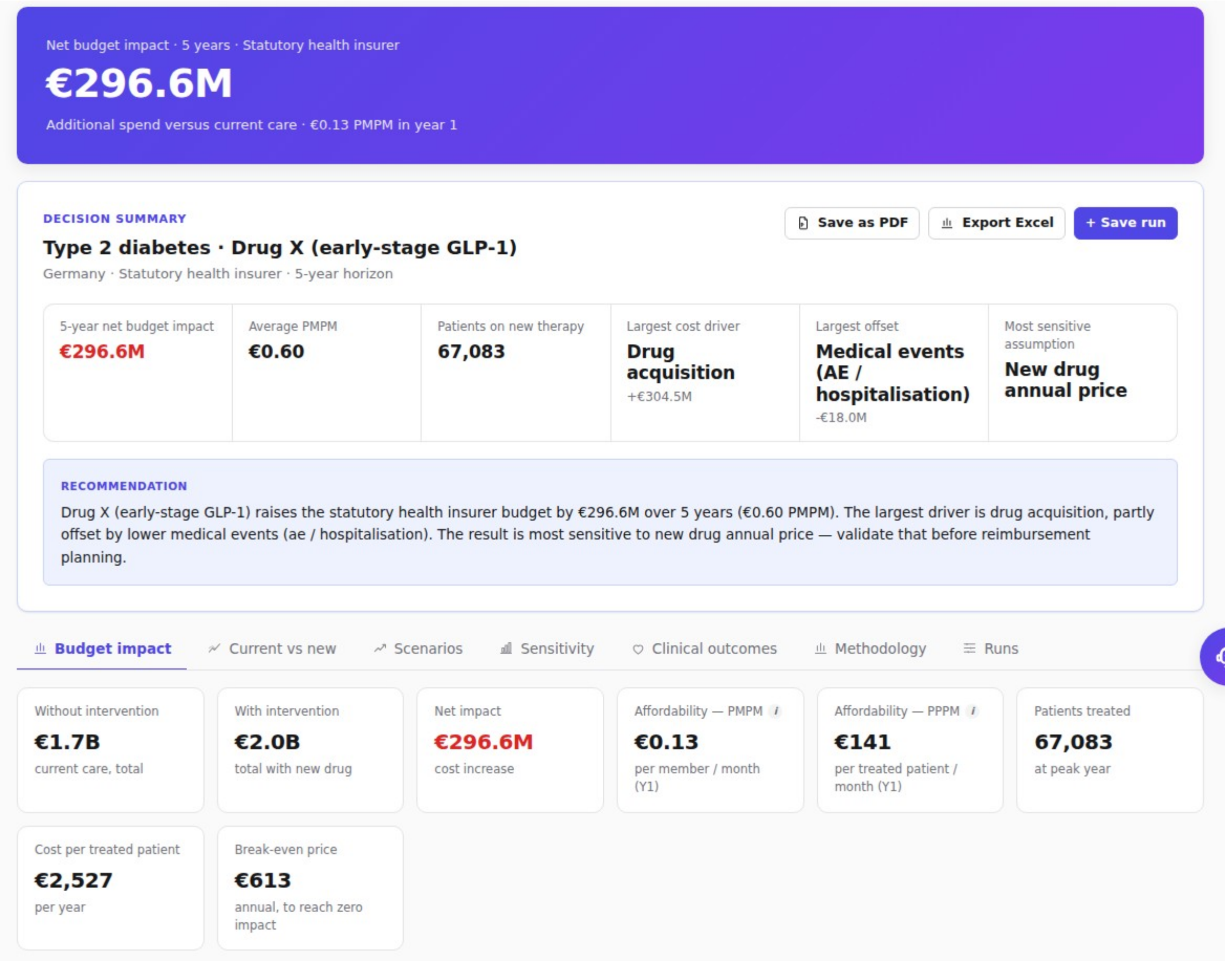
early-stage-bia.onrender.com

Code

github.com/Sumithraju/
early_stage_bia_react_node_postgres

Backup slides follow: the result as rendered, the sensitivity tornado, the audit trail, how it is tested, and what it does not do.

Results, as the tool renders them



The headline

€296.6M over five years, with €0.13 PMPM stated beside it.

A decision summary

Largest cost driver, largest offset and the most sensitive assumption, above every tab.

Affordability, two ways

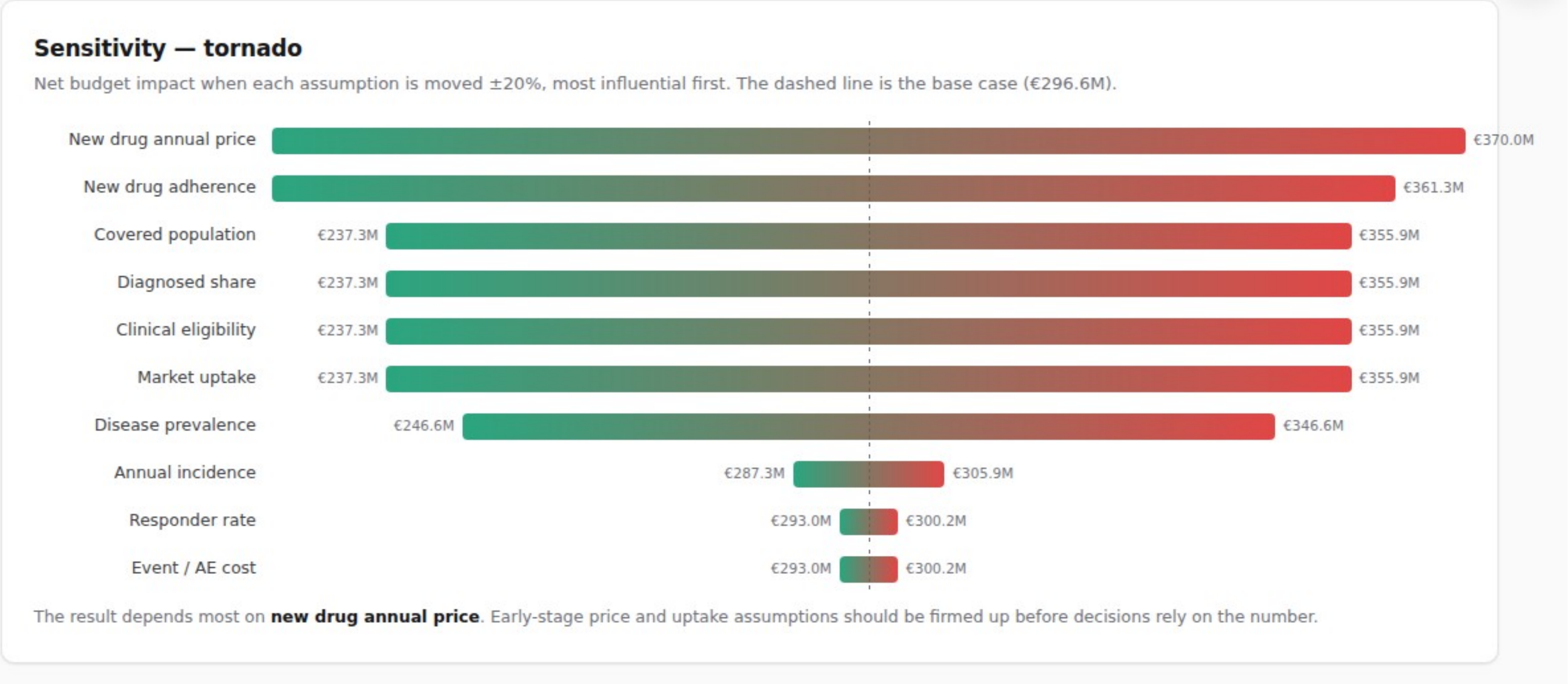
€0.13 per covered member per month; €141 per treated patient per month.

The negotiating figure

€613 is the annual price at which net impact reaches zero.

What would change the answer

Each assumption moved ±20% on its own, sorted by how far the five-year answer swings. The dashed line is the base case.



How to read it

Ten parameters, one at a time, re-running the whole engine for each.

New drug annual price ranks first, so that is where the evidence budget goes.

Price and uptake dominate. Event costs barely move the answer.

It answers a reviewer's real question: what would change your mind?

One-way sensitivity only. There is no probabilistic analysis today, which is the first item on the roadmap.

Every figure traces back to an input

The methodology view inside the app, rendered from the same result object as the headline. No reviewer has to open the code.

How the number is calculated

Every figure below comes from your inputs — no hidden steps.

Covered population

× prevalence × diagnosed × clinical × payer × access × willingness

↓

Eligible patients = 181,764 (year 1)

↓

× current-care mix and new-drug uptake each year

↓

Patients treated = 67,083 at peak

↓

each patient costs: drug + administration + monitoring + medical events − avoided events

↓

Net budget impact = (with-intervention − without-intervention) spend = €296,601,076

Covered population → eligible patients → patients treated → cost per patient → net budget impact. Five steps, no hidden ones.

Parameter provenance

The database records where each value came from, when it was retrieved, its bounds and whether a user overrode it.

Precedence, in order

User override, then scenario, then country-specific, then default assumption.

Reconciled, not approximated

Cost components sum exactly to the scenario totals. The split is the same arithmetic, not a second estimate.

How we test a number nobody can eyeball

38 unit tests run against the engine, plus browser checks against the built bundle and endpoint probes against a running server.

The answer is arithmetically right

Funnel	8,000,000 covered lives resolve to 181,764 eligible in year 1.
Components	Category costs sum to the scenario total exactly, not approximately.
Break-even	Net impact is zero at €613, found by binary search.
Monotonic	€148.3M at half uptake, €296.6M at base, €444.9M at 1.5×.
PMPM	€12,269,434 in year 1 over 8,000,000 lives over 12 is €0.13.

It fails loudly rather than quietly

Shares ≠ 1.0	Throws and names the total, rather than modelling 90% of a market.
No comparator	Throws. There is nothing to compare the intervention against.
Negative rate	Clamped to zero. No NaN propagates into a currency figure.
Zero population	Net impact is zero, with no division by zero.
No database	The server boots, health reports it, and the tool still works.

Stated plainly: the browser checks are run per change rather than committed to CI, no test covers the PostgreSQL-backed paths end to end, and live third-party calls are stubbed.

What this is not

An early-stage estimation tool, not a validated HTA submission model. Clinical, payer, epidemiological and price assumptions should be reviewed by HEOR specialists before any decision relies on them.

Single country

One market, one currency and one price set per run. Multi-country aggregation is not implemented.

Subgroups are partial

Selecting a subgroup narrows the population but inherits the parent's cost and outcome parameters.

Proportional displacement

When the new drug takes share it is drawn evenly from comparators. Real displacement rarely is.

One-way sensitivity

No probabilistic analysis and no correlated parameter sampling.

Illustrative demo values

The built-in scenario is plausible but not drawn from a validated published source.

Free-tier hosting

The instance sleeps after 15 minutes and the free database expires after 30 days, which is why the tool runs without one.